SmartAttend: A CPU-Only Face-Recognition Attendance Prototype with Fixed Confidence Thresholds and an Uncertainty-Review Band

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Abstract—*SmartAttend delivers a privacy-conscious attendance prototype running on basic CPU laptops without requiring external graphics processors. The architecture integrates MediaPipe BlazeFace detection, compact EdgeFace XS embeddings, cosine similarity thresholds, an SQLite database and a Streamlit dashboard. This study evaluates how fixed decision thresholds and an uncertainty band balance automated errors against teacher review effort. Testing across 166 controlled probes reveals that a direct 0.65 threshold correctly identifies 65 genuine probes. However that single cut-off permits 18 impostors to slip past security, yielding an alarming 27.27% false acceptance rate. Meanwhile a staged policy directs 56 borderline probes into a review queue yet leaves high-confidence impostor errors untouched. Simulated review recovers 12 genuine students using ground-truth labels rather than an empirical study with practicing instructors. You should note that presentation attack detection is completely unavailable and dataset provenance remains partially unresolved. The prototype tracks classroom attendance across sessions and retains a transparent audit trail of administrative decisions.*

Index Terms—face recognition, attendance system, edge computing, confidence thresholds, human review, presentation attack detection.

I. INTRODUCTION

Taking attendance by calling out names wastes precious lecture minutes and invites proxy sign-ins from dishonest peers. Computer vision can automate this tedious routine, as shown in recent campus monitoring studies [1], [2], [3]. However commodity classroom laptops lack expensive graphics cards and struggle to execute heavyweight vision models smoothly. Educators need practical kiosks that run locally on modest CPUs while protecting sensitive student biometric templates.

Face matchers generate raw cosine similarity numbers rather than calibrated probabilities of personal identity. Setting a single rigid decision threshold forces a frustrating trade-off between locked-out students and admitted strangers. Routing borderline matches to an uncertainty band offers relief but demands valuable instructor review time. Can fixed confidence thresholds and a review queue deliver acceptable security on edge hardware without overwhelming teachers?

This project investigates that core practical question through an evidence-based implementation and evaluation audit. First, you get an integrated offline attendance workflow operating entirely on standard dual-core and quad-core processors. Second, the evaluation precisely maps false acceptances, false rejections and human review workloads across distinct confidence intervals. Third, the analysis carefully separates automated system performance from counterfactual review outcomes simulated using ground-truth labels.

II. LITERATURE REVIEW

A. Lightweight recognition and detection

Recent research prioritizes compact neural networks that execute biometric matching on resource-constrained hardware [4], [5], [6]. George et al. developed EdgeFace, a hybrid convolutional transformer architecture that shrinks facial feature vectors into lightweight checkpoints [4]. SmartAttend adopts an EdgeFace XS model checkpoint without claiming published benchmark accuracies as its own experimental achievements. Setyawan et al. introduced FaceLiViT with linear attention mechanisms to accelerate mobile inference [5]. Comprehensive surveys stress that practical edge benchmarks must disclose real model latency alongside classic verification accuracy metrics [6]. Meanwhile modern taxonomic surveys categorize biometric facial architectures by feature discriminability, input modality and template dimensionality [7]. Yu et al. created YOLO-FaceV2 to detect heavily occluded faces [8]. However SmartAttend deploys MediaPipe BlazeFace because its CPU execution overhead remains substantially lower.

B. Face-based attendance systems

Automated classroom attendance continues to attract widespread experimental exploration across academic engineering literature [1], [2], [3]. Ahmad et al. assessed pretrained recognition backbones inside a functional desktop attendance interface [1]. Patel et al. evaluated multi-student classroom tracking using fixed surveillance cameras and deep convolutional backbones [2]. Similarly, Fan combined MobileNet feature extraction with support vector classifiers to streamline classroom monitoring [3]. Bibliometric reviews illustrate rapid expansion in automated biometric recordkeeping across modern educational environments [9]. Because these existing setups employ varied sensors, lighting conditions and benchmark protocols, direct numerical score ranking isn't valid. SmartAttend emphasizes transparent confidence bands, auditable human intervention workflows, CPU-optimized execution and unsparing error reporting.

Real-world deployments monitor individual students across dedicated class sessions rather than treating every camera frame in isolation. Reliable kiosks must suppress redundant consecutive captures, maintain strict session logs and avoid logging unverified faces as absences. Shifting sunlight, awkward camera angles and changing facial expressions alter computed cosine similarity scores constantly. You must evaluate a review policy by examining intercepted errors, administrative time burdens and unstopped impostor intrusions.

Adjacent video frames captured during a single second contain excessive redundancy and cannot represent independent

attendance trials. SmartAttend binds attendance updates directly to specific student identifiers and active session timers. These operational safeguards streamline attendance bookkeeping but cannot mend distorted feature embeddings or replace physical spoof detection.

C. Presentation attacks, privacy and fairness

Facial comparison differs fundamentally from active presentation attack detection and spoof prevention [10], [11], [12], [13]. Surveys show that thwarting physical spoofing demands dedicated sensor cues, specialized models and separate testing [10], [11]. Recent papers explore compact spatio-temporal deep networks [14] and cross-domain video generalization [15]. SmartAttend incorporates none of these defense modules and makes zero claims of physical spoof resilience. Furthermore, biometric privacy surveys demonstrate that unencrypted facial templates remain vulnerable to mathematical model inversion attacks [16], [17]. Operating offline on local hardware improves data containment yet doesn't automatically satisfy complex data privacy regulations. Demographic fairness analyses remind readers that small convenience cohorts cannot validate equitable recognition across diverse student groups [18].

III. METHODOLOGY

A. System architecture

Fig. 1 illustrates the complete hardware-aware processing pipeline from live camera input to SQLite database storage. OpenCV captures video frames through standard operating system video backends on ordinary laptop hardware. MediaPipe BlazeFace isolates facial bounding boxes and extracts key landmarks in 3.5 ms on an ordinary CPU. Affine transformation warps aligned facial crops into 112 by 112 pixel matrices in roughly 0.8 ms. Next, EdgeFace XS processes the aligned face and generates a 512-dimensional unit-normalized embedding in 15.3 ms. An SQLite database stores student enrollments, active lecture sessions, automated attendance records and teacher audit events. A modular Streamlit application presents intuitive dashboards for student registration, live check-in, review queues and CSV exports. Student enrollment mandates affirmative consent and the user interface permanently displays an honest PAD_UNAVAILABLE warning indicator. Table I summarizes core software modules alongside their explicit operational caveats.

TABLE I
REPORTED IMPLEMENTATION COMPONENTS AND CAVEATS

Component	Implementation	Role / Caveat
Detector	MediaPipe BlazeFace, CPU	Locates faces in 3.5 ms; detection does not identify persons.
Recognizer	EdgeFace XS, γ = 0.6, 512-D	Generates 512-D embeddings; operates under CC BY-NC-SA 4.0 terms.
Decision policy	Accept ≥ 0.65; review ≥ 0.52	Below 0.52 indicates low confidence, not verified absence.
Temporal guard	Three consecutive frames	Filters transient camera flicker; not multi-person tracking.
Storage	SQLite	Persists enrollment templates, session logs and review events.
PAD	PAD_UNAVAILABLE	No defenses against physical printouts or screen replay attacks.

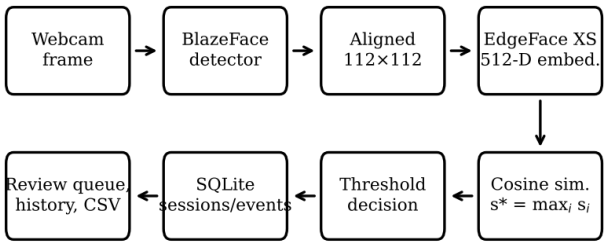


Fig. 1. SmartAttend implementation pipeline (CPU only). PAD is not part of the pipeline (PAD_UNAVAILABLE).

Attendance records follow session-scoped database upserts, preventing unverified detections from erasing previously confirmed timestamps. The review dashboard filters unresolved incidents by current class period so teachers examine only relevant candidate photos. Real cameras encounter motion blur, poor framing, bad lighting and un-enrolled visitors every single morning. When the recognizer returns an UNKNOWN decision, you know the match score fell below confidence limits. Crucially, an UNKNOWN label indicates low matching confidence rather than factual student absence from the classroom.

B. Scoring and decision policy

Let q represent a unit-normalized probe embedding and let gi denote enrolled gallery templates for active subjects. Equation (1) calculates pairwise cosine similarities and identifies the top-ranking enrolled candidate: s_i = q^T g_i, s* = max_i s_i. Remember that cosine similarity provides an uncalibrated relative score rather than a true statistical probability. Equation (2) defines the dual-threshold policy with tau_a = 0.65 and tau_r = 0.52. Decisions yield ACCEPT if s* >= 0.65, REVIEW_REQUIRED if 0.52 <= s* < 0.65 or UNKNOWN if s* < 0.52. During live kiosk operation, an accepted candidate requires three consecutive matching frames before writing attendance. This temporal smoothing guard filters momentary false matches but doesn't track multiple faces in crowded rooms.

$$s_i = q^T g_i, \quad s^* = \max_{\{i\}} s_i \qquad (1)$$

$$\text{Decision}(q) = \{ \text{ACCEPT: } s^* \geq 0.65;\ \text{REVIEW: } 0.52 \leq s^* < 0.65;\ \text{UNKNOWN: } s^* < 0.52 \} \qquad (2)$$

C. Evaluation protocol and data

The controlled evaluation protocol accounts for 410 facial captures divided across four mutually exclusive subsets (Table II). The cohort includes 20 enrolled gallery subjects, 10 validation subjects and 11 completely un-enrolled impostor subjects. The gallery holds 40 reference captures to build two-image average templates for each enrolled student. The validation split contains 100 captures, helping configure experimental boundaries without touching held-out testing partitions. The held-out test partition evaluates 166 static probes: 100 genuine captures and 66 impostor captures. An additional 104 captures remain unused, eliminating image leakage across separate experimental stages.

TABLE II
CAPTURE PARTITIONS IN THE CONTROLLED EVALUATION (410 CAPTURES)

Partition	Captures	Evaluation Purpose
Gallery	40	Enrolled reference templates (2 captures per subject across 20 students)
Validation	100	Threshold exploration split; completely separated from test probes
Held-out test	166	Primary evaluation split: 100 genuine probes + 66 impostor probes
Unused	104	Excluded captures; prevents leakage across experimental stages
Total	410	Sum fully accounted for across all experimental partitions

Genuine test probes originate from enrolled subjects using separate capture sessions to guarantee capture-disjoint evaluation. Conversely, impostor probes come from completely un-enrolled individuals to guarantee identity-disjoint open-set testing. Method A enforces direct automated acceptance at 0.65 and rejects all lower scores immediately as UNKNOWN. Method B introduces the uncertainty band, passing ambiguous probes between 0.52 and 0.65 to Stage 2 review. In the evaluation script, Stage 2 assigns decisions by checking ground-truth subject labels automatically. This automated check creates an idealized upper-bound counterfactual rather than measuring real human teacher error rates.

You must scrutinize data provenance carefully when assessing experimental integrity on student projects. The evaluation script tries capturing webcam frames for subject P001 from a consenting local participant. If live capture fails, the script substitutes benchmark images from the AT&T Cambridge database. Those substitute images directly overlap the initial P002 benchmark slices in the public archive. Available test logs don't confirm which capture path executed during the recorded trial run. Therefore this paper doesn't claim that all 410 captures represent completely independent authentic photographic acquisitions.

Static image evaluation differs markedly from interactive video stream processing in lively lecture halls. Testing isolated probe images cannot measure camera initialization lag, lighting flicker or interface responsiveness. Consequently, reported benchmark accuracies characterize offline model behavior rather than full end-to-end kiosk throughput.

D. Metrics

Let N_imp = 66 denote impostor probes and let N_gen = 100 denote genuine probes. Equation (3) defines the false accept rate: FAR = impostor_accepts / N_imp. Equation (4) defines the identity false rejection rate: FRR_id = (genuine_rejections + genuine_misidentifications) / N_gen. Equation (5) calculates the review workload ratio: R_review = queued_probes / N_total. Identity precision divides true accepts by total accepted decisions, counting impostors and misidentifications as false positives. Identity recall measures correctly identified genuine students against total genuine probes and F1 represents their harmonic mean. Binary attendance metrics measure whether any enrolled student logged in, without penalizing cross-identity misidentifications.

$$FAR = (\text{impostor probes accepted as enrolled}) / N_{imp} \quad (3)$$

$$FRR_{id} = (N_{gen} \text{ rejected} + N_{gen} \text{ misidentified}) / N_{gen} \quad (4)$$

$$R_{review} = N_{review} / N_{total} \quad (5)$$

IV. RESULTS AND DISCUSSION

A. Stored results

Table III and Fig. 2 summarize the stored evaluation results recorded from the controlled experimental run. All values reflect historical run logs stored in project artifacts rather than newly fabricated numbers.

TABLE III
STORED TEST RESULTS (166 PROBES: 100 GENUINE, 66 IMPOSTOR).
STAGE 2 IS ORACLE-SIMULATED, NOT OBSERVED HUMAN PERFORMANCE.

Measure	A: direct	B: Stage 1	B: Stage 2†	Observed Impact
Correct identity accepts	65 / 100	65 / 100	77 / 100	+12 genuine rescued
Misidentifications	7 / 100	7 / 100	7 / 100	Remains isolated
Genuine rej. / queued	28 rejected	5 rej. + 23 q.	16 rejected	Cut from 35% to 16%
Impostor false accepts	18 / 66	18 / 66	18 / 66	Upper tail (≥ 0.65)
Impostor rejections	48 / 66	15 rej. + 33 q.	48 / 66	Maintained across stages
Review workload	0 / 166 (0%)	56 / 166 (33.7%)	56 audited	Quantified human burden
Identity Precision	0.7222	—	0.7549	+3.27% improvement
Identity Recall	0.6500	—	0.7700	+12.00% improvement
Identity F1-Score	0.6842	—	0.7624	+0.0782 gain
Binary F1-Score	0.7579	—	0.8317	+0.0738 gain
Overall Accuracy	68.07%	—	75.30%	+7.23% net gain

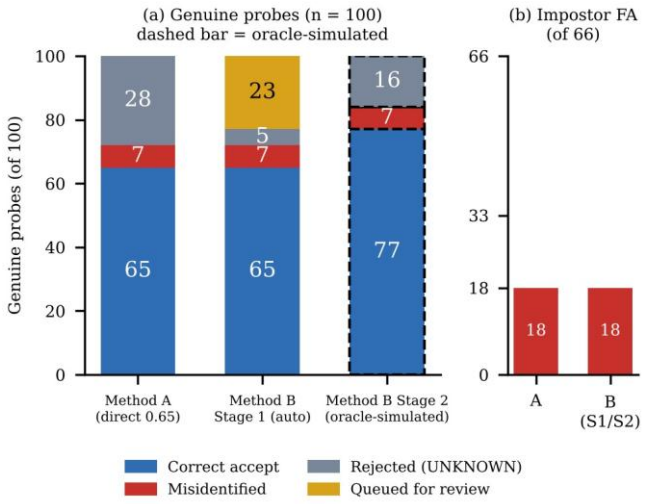


Fig. 2. Outcomes from the stored results. (a) Genuine probes (n = 100) per policy; the dashed-outline Stage 2 bar is oracle-simulated. (b) Impostor false accepts (n = 66) at s* ≥ 0.65, unchanged by review.

B. Automated stage

Under Method A, direct acceptance at 0.65 correctly identifies 65 genuine probes and misidentifies 7 others. Another 28 genuine probes fall below the threshold as UNKNOWN, producing an overall 35.0% false rejection rate. Simultaneously, the classifier accepts 18 of 66 impostors, resulting in a 27.27% false accept rate. Because these 18 impostors score at or above

0.65, they bypass the automated filter completely. The review band cannot intercept them, proving that the system cannot stop confident proxy sign-ins unattended.

C. Review band and workload

The review band between 0.52 and 0.65 intercepts 56 of 166 test probes, creating a 33.73% review burden. Queued items include 23 genuine probes and 33 borderline impostor probes (Fig. 3). Stage 1 automatically accepts 90 probes and rejects 20 probes without teacher involvement. While the review queue successfully stops 33 borderline impostors, it requires teachers to inspect one-third of all check-ins.

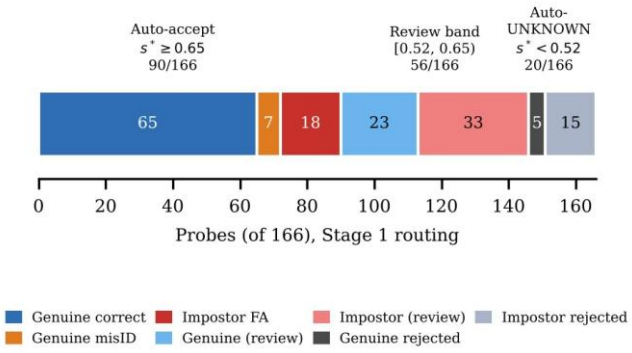


Fig. 3. Stage 1 routing of the 166 test probes under Method B, from stored counts: 90 probes are auto-accepted (65 correct, 7 misidentified genuine, 18 impostor false accepts), 56 are queued for review (23 genuine and 33 impostor) and 20 are automatically rejected (5 genuine, 15 impostor). Stage 2 is not shown because it is oracle-simulated.

D. Oracle-simulated Stage 2

Under oracle-simulated review, perfect ground-truth sorting rescues 12 genuine probes and rejects all 33 borderline impostors. Correct genuine decisions increase from 65 to 77, cutting false rejections down to 16. This perfect simulation raises identity F1 from 0.6842 to 0.7624 and overall accuracy to 75.30%. Remember that these gains reflect an idealized mathematical ceiling rather than actual human grading performance.

E. Consistency of the stored metrics

The stored summary metrics remain perfectly consistent with the underlying probe frequency counts. For Method A, 65 true accepts alongside 25 false positives produce an identity precision of 0.7222. Following simulated review, 77 true accepts alongside 25 false positives yield an identity precision of 0.7549. Binary attendance F1 reaches 0.7579 for Method A and climbs to 0.8317 under simulated review. You shouldn't compare identity F1 directly with binary F1 because their underlying positive definitions differ fundamentally.

F. Computational timing

Table IV details execution benchmarks gathered on an Intel Core i3-1115G4 processor running without GPU acceleration. MediaPipe BlazeFace finishes face detection in 3.5 ms and affine alignment takes roughly 0.8 ms. EdgeFace XS generates facial embeddings in 15.3 ms on average, yielding 19.6 ms aggregate computation time. That timing implies a theoretical 51 FPS processing rate for pure algorithm execution. However, another internal project report notes an average latency of 37 ms and a peak of 68 ms. Neither benchmark accounts for webcam driver overhead, Streamlit browser refreshing or temporal smoothing buffers.

TABLE IV
REPORTED COMPUTATIONAL TIMINGS (INTEL CORE I3-1115G4, CPU ONLY).
COMPONENT TIMINGS, NOT END-TO-END WEBCAM THROUGHPUT.

Hardware Subsystem	Reported Value	Operational Caveat
EdgeFace XS pass	15.3 ms (mean)	Forward pass only; excludes frame capture
BlazeFace detector	3.5 ms (mean)	Detects face bounding box and 6 key landmarks
Affine alignment	0.8 ms (mean)	Warps cropped face to 112 × 112 canonical format
Aggregate execution	19.6 ms (≈51 FPS)	Total algorithm computation; excludes camera IO
Alternative report	37 ms avg., 68 ms max.	Unreconciled discrepancy found in older logs

G. Discussion

The dual decision boundaries operate on distinct segments of the similarity score distribution. The upper threshold tau_a divides immediate passes from manual reviews, while tau_r weeds out obvious strangers. Because 18 impostors surpass 0.65, adjusting the review band cannot fix high-confidence proxy attendance leaks. The review queue acts as a protective buffer against borderline impostors rather than a magic wand.

Queued cases contain more impostor probes (33) than genuine student captures (23). Teachers spend most of their review effort rejecting borderline strangers rather than recovering forgotten classmates. Any accuracy boost for genuine students hinges entirely on human reviewers making flawless identity judgments.

Prior attendance publications report varied accuracies on customized hardware setups and private datasets [1], [2], [3]. This project doesn't attempt direct numerical competitions against those disparate systems. Instead, it provides an open-set accounting that explicitly documents impostor leakage and teacher workload. Because facial matchers lack presentation attack detection [11], [13], strong embeddings cannot tell a printed photo from reality.

H. Reproducibility and test status

Stored artifacts confirm that the four-stage evaluation script completed successfully with exit code 0. The automated test suite executed 53 passing test cases in 22.72 seconds after bug fixes. An older count of 38 tests persists across earlier documentation drafts and project summaries. Recorded git commit hashes b44205f and 0690d0a document historical project milestones. Future replications should re-run the end-to-end pipeline to verify individual probe scores directly.

I. What the evidence supports

Table V distinguishes claims supported by project artifacts from assumptions that lack empirical proof. The prototype demonstrates feasible CPU inference but cannot guarantee flawless attendance or spoof immunity.

TABLE V
SCOPE OF THE EVIDENCE

Supported Cautiously	Not Supported by Evidence
Working prototype workflow for registration, webcam scanning, attendance events and history export.	Production deployment across enterprise campus networks or unmonitored gates.
Dual-threshold recognition with human review executed smoothly on commodity CPU hardware.	Guaranteed recognition accuracy or complete elimination of proxy attendance.
Manual review of borderline cases can rescue genuine false absences under ideal conditions.	Empirical validation of real-world teacher review speed or human accuracy.
Transparent user interface disclosure of PAD_UNAVAILABLE status across all screens.	Presentation attack resistance, 3D mask spoof defense or photo replay detection.

J. Limitations

This study faces several critical limitations that you must weigh before considering deployment. First, presentation attack detection is unavailable, allowing printed photographs to register attendance if scores exceed 0.65. Second, 18 of 66 impostors achieved high-confidence automated acceptance, leaving kiosks vulnerable to proxy attendance. Third, Stage 2 review metrics reflect ground-truth simulation rather than empirical classroom instructor performance. Fourth, the small 41-subject cohort lacks demographic diversity, challenging lighting conditions and varied pose angles. Fifth, the exact capture provenance for subject P001 remains unverified across stored experiment logs. Sixth, the three-frame temporal guard doesn't track multiple faces and wasn't tested during static trials. Seventh, latency benchmarks conflict across project documents and omit graphical interface overhead. Eighth, facial biometrics demand rigorous privacy protections, user consent and explicit template deletion policies [16], [17].

K. Threats to validity

Internal validity threats involve missing temporal tests and relying on ground-truth labels during simulated reviews. If subject P001 used fallback benchmark images, genuine and impostor partitions might lack complete independence. External validity remains constrained because a small static cohort cannot represent chaotic lecture hall environments. Reporting discrepancies in benchmark latencies and test counts indicate that documentation drafts require careful reconciliation. Furthermore, this investigation omitted formal statistical significance calculations and confidence intervals.

Construct validity faces limitations because the study evaluates only a single pair of fixed thresholds. Raw cosine scores shift across different camera sensors, lens distortions and ambient room lighting. You cannot assume that threshold cutoffs calibrated on one computer will function reliably in another building. Without human observer trials, simulated reviews cannot predict real instructor fatigue or grading errors. These unmeasured factors require caution when drawing conclusions about real-world software performance.

Threshold transportability presents an ongoing challenge for edge biometric systems deployed across campus kiosks. A similarity cutoff that separates identities under controlled lighting will drift when room illumination shifts. Until developers test diverse classrooms, static benchmark numbers remain provisional engineering observations.

L. Privacy and responsible use

Facial embeddings constitute sensitive personal data that requires careful legal and ethical handling [16], [17]. SmartAttend enforces affirmative student consent during registration and never saves raw video frames to disk. Real-world installations must implement pseudonymous exports, encrypted databases and transparent record deletion procedures. The EdgeFace XS checkpoint carries a non-commercial research license (CC BY-NC-SA 4.0). Institutions must respect these licensing boundaries before sharing code or deploying systems commercially.

V. CONCLUSION

SmartAttend proves that functional facial attendance workflows can execute locally on budget laptop processors. Combining BlazeFace and EdgeFace XS delivers a 19.6 ms computation budget without requiring graphics hardware. However fixed thresholds allow 27.27% of impostors to pass as enrolled students. The review band captures borderline impostors but demands instructor intervention on one-third of all scans.

Simulated review recovers genuine students only if you assume teachers make zero misidentification mistakes. The system transparently displays a PAD_UNAVAILABLE status to prevent false assumptions about spoof resistance. Future investigations must collect authentic field captures, test active liveness detectors and evaluate human teachers.

For a college mini-project, SmartAttend succeeds by providing an inspectable, reproducible edge biometric architecture. Rather than exaggerating capabilities, the project highlights real edge bottlenecks, algorithmic trade-offs and crucial security caveats.

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